

https://docs.google.com/spreadsheets/d/1JPjysrO5bZ8gPr			Number of Course Outcomes	Number of Program Outcomes	Program Outcomes (POs)											Knowledge Profile								Complex Eng. Problems (EP)							Complex Eng. Activities (EA)							
					a	b	c	d	e	f	g	h	i	j	k	l	K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5		
					Engineering Knowledge	Problem Analysis	Design of solutions	Investigation	Modern tool usage	Engineering and society	Environment and sustainability	Ethics	Teamwork and leadership	Communication	Project management and Finance	Life-long learning	Natural Sciences	Mathematics	Engineering Fundamentals	Engineering specialist knowledge	Engineering design	Engineering practice	Comprehension	Research literature	Depth of knowledge required	Range of conflicting requirements	Depth of analysis required	Familiarity of issues	Extent of applicable codes	Extent of stakeholder involvement	Interdependence	Range of resource	Level of interaction	Innovation	Engagement for society and the environment	Familiarity		
					C			CP		CA		A		PA	CA	A	Engineering		Engineering		Engineering		Engineering		Engineering		Engineering		Engineering		Engineering		Engineering		Engineering			
					K1 - K4			K5	K8	K6	K7																											
Include?	Course Code	Course Title			All POs are attained											All KPs are attained								All EPs are attained							All EAs are attained							
					X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X			
Yes	CSE400	Thesis / Project / Internship	14	11		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Yes	CSE110	Programming Language I	4	3		X	X	X											X																			
Yes	CSE111	Programming Language II	4	2	X	X													X																			
Yes	CSE220	Data Structures	5	2	X		X											X	X	X	X	X																
Yes	CSE221	Algorithms	4	3	X	X			X								X	X	X			X																
Yes	CSE230	Discrete Mathematics	7	2	X	X											X	X	X	X																		
Yes	CSE250	Circuits and Electronics	6	5	X	X	X						X	X			X	X	X			X																
Yes	CSE251	Electronic Devices and Circuits	6	5	X	X	X						X	X			X	X	X		X																	
Yes	CSE260	Digital Logic Design	5	4	X		X		X					X			X				X	X																
Yes	CSE320	Data Communications	4	2	X	X													X	X																		
Yes	CSE321	Operating Systems	3	3	X	X			X										X	X		X																
Yes	CSE330	Numerical Methods	3	2	X	X											X	X	X		X		X		X													
Yes	CSE331	Automata and Computability	5	3	X	X	X												X	X																		
Yes	CSE340	Computer Architecture	5	3	X	X					X								X	X			X															
Yes	CSE341	Microprocessors	4	2	X								X						X	X																		
Yes	CSE350	Digital Electronics and Pulse Techniques	6	6	X	X	X		X				X	X					X	X	X	X			X	X	X	X			X			X				
Yes	CSE360	Computer Interfacing	7	5	X	X	X				X			X					X		X		X		X	X	X	X	X	X	X	X	X	X				
Yes	CSE370	Database Systems	6	4	X		X		X				X				X	X	X	X		X		X	X	X	X			X								
Yes	CSE420	Compiler Design	5	4	X		X	X	X																													
Yes	CSE421	Computer Networks	5	5	X	X	X		X			X							X	X	X	X	X		X	X	X	X	X	X								
Yes	CSE422	Artificial Intelligence	6	6			X	X	X	X	X	X									X	X	X	X	X	X	X	X	X	X								
Yes	CSE423	Computer Graphics	3	2	X				X										X	X		X			X	X	X	X	X	X								
Yes	CSE460	VLSI Design	4	4		X	X	X	X																													
Yes	CSE461	Introduction to Robotics	6	6	X		X		X	X	X					X			X	X		X	X		X	X	X	X	X	X								
Yes	CSE470	Software Engineering	6	6	X		X	X					X		X	X			X	X	X				X		X	X										
Yes	CSE471	System Analysis and Design	6	6	X		X		X				X	X		X	X			X	X				X	X	X		X	X	X	X	X	X	X			

Assessment Type	CO	Assessment #	POs	
Assignment	CO1	1	a	Affective/ Characterisation
Final	CO2	2	b	Affective/ Organisation
Midterm	CO3	3	c	Affective/ Receiving
Presentation	CO4	4	d	Affective/ Responding
Project	CO5	5	e	Affective/ Valuing
Quiz	CO6	6	f	Cognitive/ Analyze
Term Paper	CO7	7	g	Cognitive/ Apply
Viva	CO8	8	h	Cognitive/ Create
Pop Quiz		9	i	Cognitive/ Evaluate
Lab Report		10	j	Cognitive/ Knowledge
Lab Test			k	Cognitive/ Understand
Lab Performance			l	Psychomotor/ Articulation
Research Report				Psychomotor/ Imitation
Project Management tools				Psychomotor/ Manipulation
Project Simulation				Psychomotor/ Naturalization
				Psychomotor/ Precision

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Sl.	CO Description	Weight: 100%	POs	Bloom's Taxonomy Domain/ Level	Knowledge Profile (K1-K8)	Complex Engineering Problems (P1-P7)	Complex Engineering Activities (A1-A5)	Comments
CO1	Understand the concepts of processes and their management, CPU scheduling techniques, and Apply system calls for solving problems on process management, and Analyze security, and protection of OS.	30	a	Cognitive/ Analyze				
CO2	Understand the behaviors of threads and their management, process, and thread synchronization mechanisms, deadlocks, virtual machines and containers and Apply thread libraries for solving problems on thread management and synchronization.	35	b	Cognitive/ Analyze				
CO3	Understand mechanisms of main memory, and virtual memory management, and file systems concepts, and implementations of OS, and Apply the concepts to design complex file systems.	35	e	Cognitive/ Apply				
	Need to discuss with course faculties							

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Sl.	CO Description	Weight: 100%	POs	Bloom's Taxonomy Domain/ Level	KPIs of relevant PO	Knowledge Profile (K1-K8)	Complex Engineering Problems (P1- P7)	Complex Engineering Activities (A1-A5)
CO1	Formulate a modern-day complex engineering and/or computing problem preferably relevant to the field.	5	b	Cognitive/ Analyze				
CO2	Identify the objectives, specifications, functional and non-functional requirements, and constraints as well as applicable compliance, standards and codes of practice to the solution of the problem	5	c	Cognitive/ Analyze				
CO3	Assess the impact of the solution of the problem in terms of societal, health, safety, legal and cultural context	5	f	Cognitive/ Evaluate				
CO4	Evaluate the sustainability and impact of solution of the proposed problem in terms of environmental consideration	5	g	Cognitive/ Evaluate				
CO5	Design multiple engineering/theoretical solutions/proof of the problem to meet the desired objectives, need and requirements within the given constraints.	15	c	Cognitive/ Create				
CO6	Analyze and assess alternative design solutions for the problem to determine the most appropriate one, considering factors such as cost, efficiency, usability, manufacturability, impact, sustainability, maintainability, and performance in relation to provided specifications, prerequisites, and benchmarks.	10	d	Cognitive/ Evaluate				
CO7	Complete the final design and development of the solution with necessary adjustment based on performance evaluation	5	c	Cognitive/ Create				
CO8	Use contemporary engineering and information technology tools for the design, creation, and validation of the work.	10	e	Cognitive/ Apply				
CO9	Conduct independent research, literature survey and learning of new technologies and concepts as appropriate to identify, analyze, and formulate the problem as well as design, develop and validate the solution.	5	l	Cognitive/ Create				
CO10	Demonstrate resource management skill in various stages of developing the solution to the problem.	3	k	Affective/ Valuing				
CO11	Perform economic analysis of the resource management.	2	k	Cognitive/ Evaluate				A1
CO12	Incorporate ethical considerations and uphold academic responsibilities during the solution design process and throughout all phases of the work.	5	h	Affective/ Characterisation				A4
CO13	Function effectively as an individual and as a team member for successfully completion of the work.	5	i	Affective/ Valuing				A2
CO14	Demonstrate effective communication by means of written documents, journals, technical reports, deliverables, presentations, and verbal exchanges as deemed suitable throughout the various stages of the work	20	j	Affective/ Organisation				